

“Math is Cool” Masters -- 2025-26

5th Grade

Mental Math Solutions

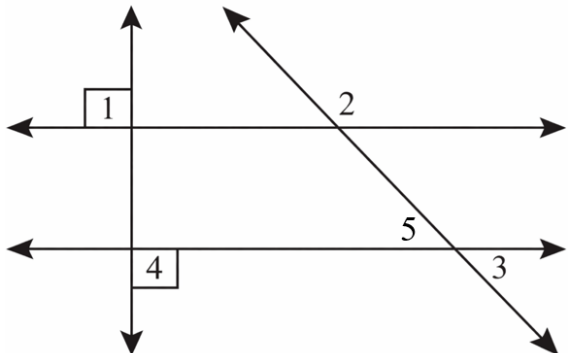
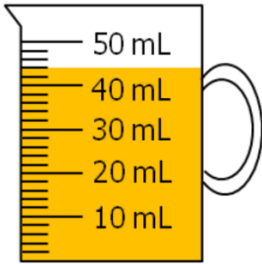
	Answer	Solution
1	12 [meters]	In meters, what is the perimeter of a square with an area of nine square meters? $9 = 3 \times 3$, $3 \times 4 = 12$
2	300 [calories]	Playing ping pong for one hour burns one hundred fifty calories. How many calories would you burn playing ping pong for one hundred twenty minutes? $120 \text{ minutes} = 2 \text{ hours}$, $150 \times 2 = 300$
3	16 [= range]	What is the range of the following set: twelve, seventeen, fifteen, twenty-eight $28 - 12 = 16$
4	5	What is the third smallest positive integer factor of one hundred five? The three smallest factors are 1, 3, and 5
5	6 [ways]	A hot air balloon took off with a sheep, a duck, and a rooster on board. However, they had to board one at a time. In how many different orders could the animals have boarded the balloon? $3! = 6$
6	3 [=x]	Solve this equation for x: Three X plus two X plus five plus seven equals twenty-seven $3x + 2x + 5 + 7 = 5x + 12 = 27$ $5x = 15$ $x = 3$
7	72 [dogs and cats]	The ratio of dogs to cats at an animal shelter was seven to five. After twelve dogs and zero cats were adopted out, the ratio of dogs to cats is now one to one. How many total dogs and cats were at the shelter originally? $D:C = 7:5 = 42:30$ After 12 dogs adopted, $D:C = 30:30$. Total originally = $42 + 30 = 72$
8	8800 [yards]	A lion's roar can be heard from five miles away. From how many yards away can a lion's roar be heard? $5 \times 1760 = 8800$

“Math is Cool” Masters -- 2025-26

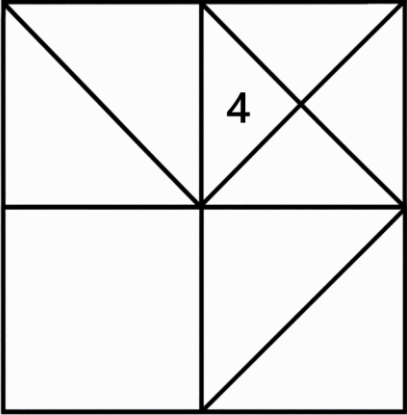
5th Grade

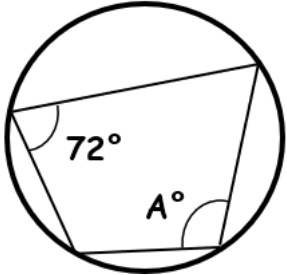
Individual Test Solutions

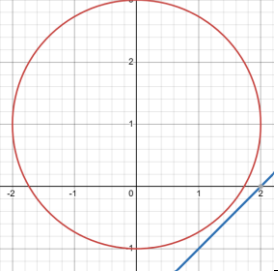
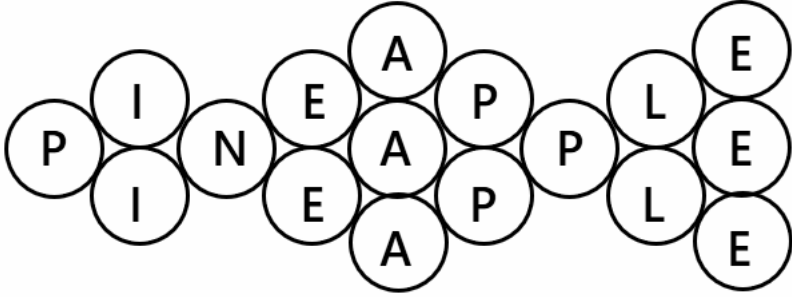
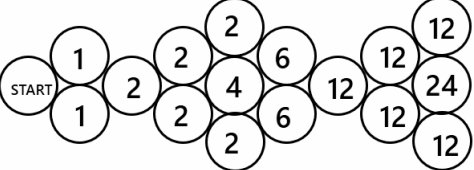
	Answer	Solution
1	2	What is the smallest positive even integer? 2 is the smallest positive even integer
2	54	What is the next number in the sequence 2, 6, 18, _ $18 \times 3 = 54$
3	3 [yards]	Nine feet is equal to how many yards? $9/3 = 3$
4	8 [zeroes]	The Earth is approximately 4.5 billion years old. When this number is written out as an integer, how many zeros are at the end of the number? 4,500,000,000 has 8 0s
5	14 [people]	There are 6 people ahead of Xinyin in a line, and 6 people behind the person directly behind Xinyin in the same line. How many total people are in the line? $6 + 7 + 1 = 14$
6	90 [%]	Ten percent of people are left-handed. If one person is randomly selected, what is the probability in percent that they are not left-handed? 10 % left-handed means 90% not left-handed.
7	25 [%]	Pound cakes got their name because the original recipe contained a pound each of butter, sugar, flour, and eggs. What percent, by weight, of a pound cake is flour? The four ingredients are in equal parts, and $\frac{1}{4} = 25\%$
8	100 [minutes]	There are about 1800 steps to the top of the Eiffel tower. If Jorge walks up them at a constant rate of 18 steps per minute, how many minutes would it take for him to get to the top? $1800/18 = 100$
9	220 [million dollars]	Queen Elizabeth II had a personal net worth of \$425 million, which included the \$65 million Sandringham House and the \$140 million Balmoral Castle. How many million dollars of her net worth was not part of these two castles? $425 - 65 - 140 = 220$
10	121	What is 11^2 ? $11 \times 11 = 121$

11	2 [angle #]	Which angle is obtuse? Your answer should be the angle number: 1, 2, 3, 4 or 5 Angles 1 and 4 are right. Angles 3 and 5 are acute. Angle 2 is obtuse. 
12	2127 [year]	Santa Claus was issued a pilot's license by the U.S. government in 1927! However, he does have to conduct a pilot review every two years for the rest of eternity, starting in 1929. In what year will he have his 100th pilot review? $1927 + 2(100) = 2127$
13	44 [mL]	How many milliliters (mL) of liquid are contained in this measuring cup? Each small line = 2 mL 
14	60 [square meters]	What is the area of a parallelogram, in square meters, with base length 12 meters and height 5 meters? $5 \times 12 = 60$
15	6580	Evaluate: $7345 - 765$ $7345 - 765 = 6580$
16	15 [years]	A Statistics class has twenty 15-year-olds, nineteen 14-year-olds, and one 34-year-old (the teacher). What is the mode, in years, of the ages of the people in this class? 15 is the most common age

17	4	A piece of paper is folded in half, and then a hole is punched through it. After the paper is unfolded again, which image number will it look like? Your answer should be the image number: 1, 2, 3 or 4. When the paper is unfolded, the hole at the bottom will also transfer up to a hole at the top.
18	840 [languages]	About 7000 languages are spoken in the world. Approximately 12% of the world's total languages are spoken in Papua New Guinea. How many languages are spoken in Papua New Guinea? $7000 \times 0.12 = 840$
19	5	What number, when multiplied by 6, is equal to 30? $5 \times 6 + 2 = 32$
20	2	In this number pattern, what number should replace the question mark? Going diagonally from top to bottom in each circle, the numbers go down by 1. $3 - 1 = 2$
21	16	Evaluate: $2\left(\frac{18}{4+5} + 3\right) + 6$ $2\left(\frac{18}{4+5} + 3\right) + 6 =$ $2(2 + 3) + 6 = 16$
22	446 [deg F]	Cooking oil boils at 230 degrees Celsius (C). Degrees Fahrenheit (F) can be found from degrees C as follows. What is the boiling temperature of cooking oil in degrees F? $F = \frac{9}{5}C + 32$ $F = \frac{9}{5}(230) + 32 = 446$

23	64 [sq. units]	A large square is divided into 4 smaller squares, and then some diagonals are drawn. One small triangular shape has an area of 4 square units. What is the total area of the large square in square units? Each small square = $4 \times 4 = 16$ $16 \times 4 = 64$ 
24	47 [day no.]	The world record for stuffing drinking straws into your mouth at once is 650. Emilia really wants to break this record, so she starts training. On the first day, she manages to cram 100 drinking straws in her mouth, and every day after that, she increases the number of straws by 12. On what number day will she break the record? $100 + 12(46) = 652$
25	90 [degrees]	A triangle has side lengths 6, 8, and 10. What is the degree measure of its largest angle? This is basically just a 3-4-5 right triangle
26	24 [ways]	In a round of musical chairs, there are 4 people and 3 chairs remaining, colored red, green, and blue. How many ways are there for people to sit down at the end of the round? $4! = 24$
27	15 [years]	A Statistics class has twenty 15-year-olds, nineteen 14-year-olds, and one 34-year-old (the teacher). What is the mean number of years of the ages of the people in this class? $(20)(15) + 19(14) + 34 / 40 = 15$

28	100 [panda-shaped solar farms]	China has made a number panda-shaped solar farms in order to get more young people like you interested in renewable energy! When a certain number of solar panels are arranged in rows of 7 per row, there are 2 left over. When arranged in rows of 9 solar panels per row, there is 1 left over. When arranged in rows of 10 solar panels per row, there are none left over. What is the fewest number of solar panels in this group? The multiples of 10 that are one more than a multiple of 9 are 10, 100, 190... 10 is not 2 more than a multiple of 7. $100 = 7(14) + 2$																																																																						
29	12 [likes]	Chloe got 3 likes on her TikTok video, Adalynn got 1 like on her TikTok video, and Yesenia got 25 likes on her TikTok video. How many likes did Luca get on his TikTok video? Number corresponds to the letter of the alphabet.																																																																						
30	406 [pillows]	May 29 is "Put a Pillow on Your Fridge Day." To prepare, Wrenly buys one pillow on May 1, two pillows on May 2, continuing this pattern until she buys 28 pillows on May 28. On May 29, instead on buying any more pillows, she puts them all on her fridge. How many pillows does Wrenly put on her fridge? $(28/2)(29) = 14 \cdot 29 = 406$																																																																						
31	10235	What is the smallest 5-digit positive integer in which three of the digits are distinct (different) odd integers and the other two digits are distinct even integers? Three smallest odd digits are 1, 3, 5. Smallest even digits are 0, 2. Start with smallest non-zero integer and then go in order from smallest to largest: 10235.																																																																						
32	2 [times]	How many times does the number 842 appear in the figure, when read from left to right with the digits in consecutive order? Find the number 842 <table><tr><td>2</td><td>7</td><td>8</td><td>4</td><td>4</td><td>2</td><td>2</td><td>3</td><td>8</td><td>7</td></tr><tr><td>0</td><td>2</td><td>0</td><td>3</td><td>9</td><td>5</td><td>4</td><td>4</td><td>6</td><td>2</td></tr><tr><td>7</td><td>6</td><td>0</td><td>6</td><td>3</td><td>8</td><td>4</td><td>9</td><td>8</td><td>3</td></tr><tr><td>5</td><td>3</td><td>6</td><td>3</td><td>3</td><td>7</td><td>4</td><td>1</td><td>6</td><td>9</td></tr><tr><td>6</td><td>0</td><td>9</td><td>8</td><td>8</td><td>0</td><td>8</td><td>4</td><td>2</td><td>3</td></tr><tr><td>3</td><td>6</td><td>5</td><td>9</td><td>6</td><td>1</td><td>9</td><td>7</td><td>9</td><td>9</td></tr><tr><td>8</td><td>4</td><td>1</td><td>5</td><td>9</td><td>2</td><td>5</td><td>9</td><td>2</td><td>1</td></tr></table>	2	7	8	4	4	2	2	3	8	7	0	2	0	3	9	5	4	4	6	2	7	6	0	6	3	8	4	9	8	3	5	3	6	3	3	7	4	1	6	9	6	0	9	8	8	0	8	4	2	3	3	6	5	9	6	1	9	7	9	9	8	4	1	5	9	2	5	9	2	1
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33	108 [degrees]	A quadrilateral is inscribed in a circle. What is the value of angle A in degrees? This is a cyclic quadrilateral, so $180 - 72 = 108$ 																																																																						

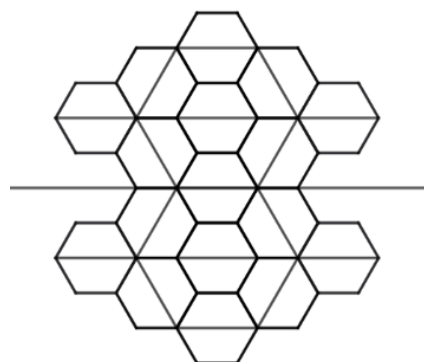
34	70	Starting from the third term, a sequence follows the formula $a_n = 2a_{n-1} + a_{n-2}$, where a_n is the nth term. In other words, every term, starting from term 3, is equal to twice the term before it added to the term before that. If the sequence starts 0, 1,..., what is the 7th term? $2(1) + 0 = 2$ $2(2) + 1 = 5$ $2(5) + 2 = 12$ $2(12) + 5 = 29$ $2(29) + 12 = 70$
35	0 [points]	On the coordinate plane, a circle with radius 2 is centered at the point (0, 1). In how many points will it be intersected by the line $y = x - 2$? 
36	48 [ways]	In how many different ways can the word PINEAPPLE be read from this diagram, starting at the left-hand P, and only moving to the right, to circles that are tangent to (touching) the current circle.  At each new circle, indicate how many ways there are to get there. At the end, add up $12 + 24 + 12 = 48$. 
37	22 [times]	How many times do the minute and hour hand overlap on an analog clock in one day, starting at midnight (12:00 AM) and ending at 11:59 PM that same day? The minute hand laps the hour hand 11 times in 12 hours, thus 22 times in 24 hours. These can be seen as happening at 12:00, ~1:05..., however, during the 11-12 period, the merge happens at 12:00, no time before that, so it's 11 times in every 12 hours instead of 12.

38	14 [= 100th term]	The first 6 terms of an infinite sequence are 1, 2, 2, 3, 3, 3, and so on, where each number is repeated the number of times equal to its value. What is the 100th term in this sequence? If n represents the value of the largest term in the sequence, then $n(n + 1)/2 = 91$ total terms when $n = 13$. Therefore, the next 14 terms will be 14, so the 100th term is 14. This one could definitely also be brute-forced by writing out the numbers.
39	16	Julius built a neural network to recognize handwritten digits (the numbers 0 to 9). For each input handwritten digit, it outputs a digit as a guess. However, there are a few problems. Any number with a "closed loop" (4, 6, 8, 9, and 0) gives an output that is that number subtracted from 9 (for example, 4 results in an output of 5). Any number that has an open loop (2, 3, 5, and 7) results in an output that is that number subtracted from 7 (for example, 3 results in an output of 4). The number 1 always results in an output of 1. If Julius inputs two handwritten digits into the neural network in order to get a two-digit output that does not start with 0, what is the smallest number that this output could <u>not</u> be? Here are the various inputs/outputs: 0 → 9 1 → 1 2 → 5 3 → 4 4 → 5 5 → 2 6 → 3 7 → 0 8 → 1 9 → 0 6, 7, and 8 cannot be outputs, and the two digit output can't start with 0, so the answer is 16

40

5 [white pairs]

Each half of this figure is composed of 4 red trapezoids, 6 blue trapezoids and 7 white trapezoids. When the upper half is folded down over the centerline, 2 pairs of red trapezoids coincide, as do 4 pairs of blue trapezoids. There are 2 blue-white pairs. How many white pairs coincide?



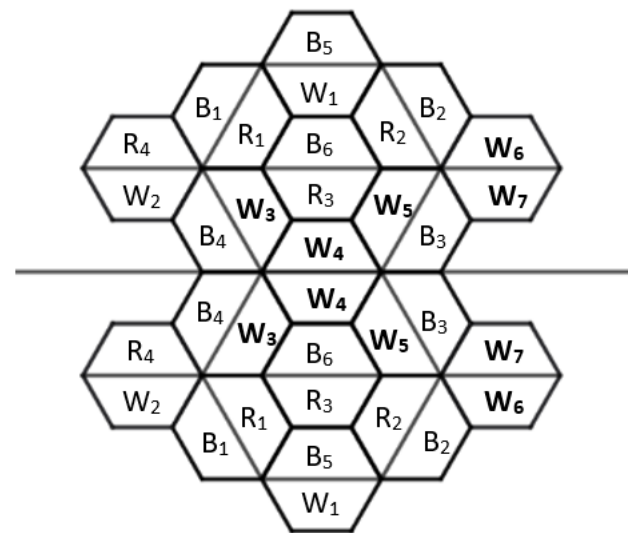
There are 4 coinciding pairs of blue trapezoids, labeled in the diagram below B1, B2, B3, and B4, in both the top and the bottom half.

The 2 blue-white pairs have a blue from the top half coinciding with a white from the bottom half, labeled B5 and W1, and a white from the top half coinciding with a blue from the bottom half labeled W1 and B5. This makes a total of 5 blue trapezoids in each half.

Therefore, there must be two blue-red pairs, a blue from the top half coinciding with a red from the bottom half, labeled B6 and R3, and a red from the top half coinciding with a blue from the bottom half labeled R3 and B6.

This makes a total of 3 reds in each half, so there must still be 2 red-white pairs, a red from the top half coinciding with a white from the bottom half, labeled R4 and W2, and a white from the top half coinciding with a red from the bottom half labeled W2 and R4.

This leaves room for 5 pairs of coinciding white trapezoids, and these are labeled with bolded W3, W4, W5, W6, and W7, in both the top and the bottom half. So, the final answer is 5.



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5th Grade

Multiple Choice Solutions

	Answer	Solution																																																																								
USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #1 THROUGH #4.																																																																										
Each cell of this 6x6 grid contains an integer. Some of these integers are not revealed to you, and a couple of them are indicated by symbols. The numbers in each column and each row form an arithmetic sequence. An arithmetic sequence is a list of numbers where the same number is added or subtracted each time to get the next number. For example, 2, 5, 8, 11, 14 is an arithmetic sequence because the number 3 is being added each time.																																																																										
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1	A	What is the value of the integer marked with a "?" A) -14 B) -16 C) -18 D) -20 E) Answer not given. Subtract 3 each time: 1, -2, -5, -8, -11, -14																																																																								
2	B	What is the mean (average) of all the integers in the third column from the left? A) 6 B) 10.5 C) 12 D) 12.5 E) Answer not given. 23, 18, 13, 8, 3, -2 Sum is 63 63/6 = 10.5																																																																								

3	D	What is the sum of all the integers marked in the four shaded cells? A) 64 B) 71 C) 75 D) 79 E) Answer not given. By filling out all the arithmetic sequences, you should get 12, 8, 28, 31. The sum is 79
4	B	In the grid above, 6 of the integers were revealed. What is the fewest number of integers that could be revealed where a student could still determine the identity of every other integer in the grid? A) 3 B) 4 C) 5 D) 6 E) Answer not given. You would need two values to find a single row, another to find an extra column, and the final one to figure out the rest of the grid.

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #5 THROUGH #7.

The sport of figure skating has jumps that are scored with the following table:

Note: Abbreviations for the jumps are written with the number of revolutions first, then the jump type. For example, 3T would be a triple toe loop with 3 revolutions and 4A would be a quadruple axel with 4 revolutions.

POINTS FOR EACH JUMP, BY NUMBER OF REVOLUTIONS

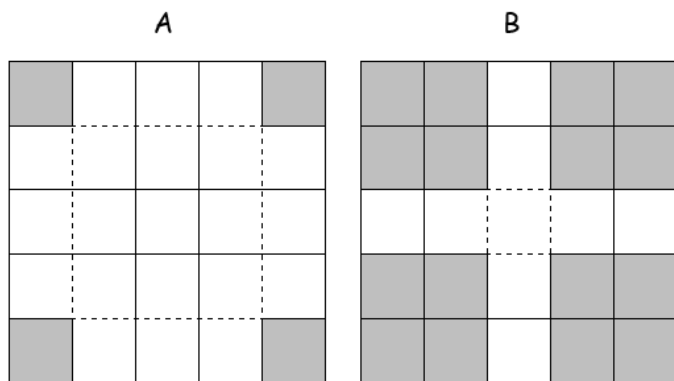
JUMP TYPE	Single (1)	Double (2)	Triple (3)	Quadruple (4)
TOE LOOP (T)	0.40	1.30	4.20	9.50
SALCHOW (S)	0.40	1.30	4.30	9.70
LOOP (L)	0.50	1.70	4.90	10.50
FLIP (F)	0.50	1.80	5.30	11.00
LUTZ (LZ)	0.60	2.10	5.90	11.50
AXEL (A)	1.10	3.30	8.00	12.50

5	C	If Ilia does two triple flips, three triple toe loops, one quadruple lutz and one quadruple axel in his long program, how many total revolutions has he made? A) 18 B) 21 C) 23 D) 24 E) Answer not given. $2 \times 3F = 6$ $3 \times 3T = 9$ $1 \times 4L = 4$ $1 \times 4A = 4$ Total = 23 revolutions
6	E [10.80]	A common Olympic-level jump combination is a triple flip (3F) followed by a triple toe loop (3T) followed by a double toe loop (2T). What is the total value of this combination, using the values in the table above? A) 10.40 B) 10.50 C) 10.60 D) 10.90 E) Answer not given. $5.30 + 4.20 + 1.30 = 10.80$
7	B	A single Euler is another type of jump that can create combinations by coming in between two different jumps. How many possible three-jump combinations with three different jumps have the single Euler as the second jump? Jumps are different if they either have a different type or a different number of revolutions. For example, a double axel (2A) is different from a triple axel (3A). A) 529 B) 552 C) 576 D) 1656 E) Answer not given. $24 \times 1 \times 23 = 552$ There are 24 total jumps in the table, so 24 ways to pick the first jump, 1 way to pick the second (single Euler) and 23 different ways to pick the third.

USE THE FOLLOWING INFORMATION TO SOLVE PROBLEMS #8 THROUGH #10.

A square grid composed of unit squares (side length = 1 unit) can be used to construct different sized boxes, with a square base and no top, by removing squares from each corner of the grid and folding up the sides.

For example, in a 5x5 square grid, either 1x1 squares can be cut out of each corner (A), or 2x2 squares can be cut out of each corner (B). In each case, the remaining sides can be folded upwards along the dotted lines to create a box.



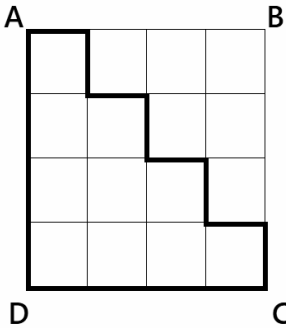
8	B	In example B shown above, what is the volume in cubic units of the box created when the four sides are folded up? A) 1 cubic units B) 2 cubic units C) 4 cubic units D) 5 cubic units E) None of the above. The base is 1x1 unit. The height is 2 units. Therefore, the volume will be $1 \times 1 \times 2 = 2$ cubic units.
9	C	Instead of starting with a 5x5 grid as shown in the example, start with an 18x18 grid composed of unit squares. By cutting squares out of the corners of the 18x18 grid and folding up the sides, how many different sized boxes can be constructed? A) 6 B) 7 C) 8 D) 9 E) None of the above. The biggest square that can be cut out of each corner is 8x8, therefore they can go from 1x1 to 8x8, or 8 total.
10	C	What is the largest volume of a box that can be created in this manner by starting from an 18x18 grid? A) 392 cubic units B) 400 cubic units C) 432 cubic units D) 452 cubic units E) None of the above. One can test a few cases to find that the maximum volume will occur when a 3x3 square is cut out of each corner. This will result in a volume of: $12 \times 12 \times 3 = 432$.

"Math is Cool" Masters -- 2025-26

5th Grade

Team Test Solutions

	Answer	Solution
1	162 [minutes]	Finley is supposed to get off work at 5 P.M., but she works overtime until 7:42 P.M. that day. How many extra minutes did Finley work? $2(60) + 42 = 162$
2	8 [units]	How many units away from the y-axis is point C? C is at (8, 8), 8 units to the right of the y-axis
3	3 [zeroes]	When the following multiplication is done, how many zeros are at the end of the number? $100 \times 5 \times 3 \times 2 \times 1$ $100 \times 5 \times 3 \times 2 \times 1 = 3000$

4	32 [units]	Square ABCD is made up of smaller congruent (equal) squares. Square ABCD has a perimeter of 32 units. In units, what is the perimeter of the portion of the square with the darkened outline?  If the square has a perimeter of 32, then each side is $32/4 = 8$. Therefore each small square has a side of $8/4 = 2$. The darkened outline has 16 segments. $16 \times 2 = 32$. Alternatively, notice that it will be exactly the same as the outer perimeter of the big square.
5	326 [cents]	Biff has 2 one-dollar bills, 3 quarters, 2 dimes and 6 pennies. Then Eho gives him 5 nickels. How many total cents does Biff have now? $2 \times 100 = 200$ $3 \times 25 = 75$ $2 \times 10 = 20$ $6 \times 1 = 6$ $5 \times 5 = 25$ $200 + 75 + 20 + 6 + 25 = 326$
6	76 [points]	In this year's Science Bowl competition, Pullman High School played Moscow High School 3 times. The final scores in points were 138 - 20, 104 - 48, and 76 - 22, with Pullman winning all 3. In points, what was Pullman's average margin of victory against Moscow over the three rounds? $(138 - 20 + 104 - 48 + 76 - 22)/3 = 76$
7	1836 [= pin number]	Camden's pin number is a 4-digit positive integer that is less than 5000. It contains the digits 1, 3, 6 and 8 in some order. The 1 is not next to the 3, the 6 is not next to the 1, and the 3 is next to the 8. What is Camden's pin number? It has to contain 38 or 83. The 1 and 6 must be on either side, because they are not next to each other. It is either 1836 or 6381, but it is less than 5000, so it must be 1836.

10

33

Morse code uses a series of dots and dashes to represent each letter of the alphabet. There are two numbers spelled out in Morse code below that have been written without putting any spaces between the letters. What is the value of the larger number minus the smaller number?

Number 1 = 46

Number 2 = 13

$46 - 13 = 33$

A ●—	J ●— — —	S ● ● ●
B — ● ● ●	K — ● —	T —
C — ● — ●	L ● — ● ●	U ● ● —
D — ● ●	M — —	V ● ● ● —
E ●	N — ●	W ● — —
F ● ● — ●	O — — —	X — ● ● —
G — — ●	P ● — — ●	Y — ● — —
H ● ● ● ●	Q — — ● —	Z — — ● ●
I ● ●	R ● — ●	

Number 1:

● ● — ● — — — ● — ● — — ● — — ● ● ● ● — ● ● —

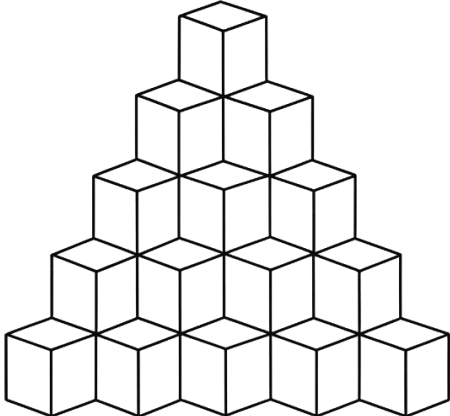
Number 2:

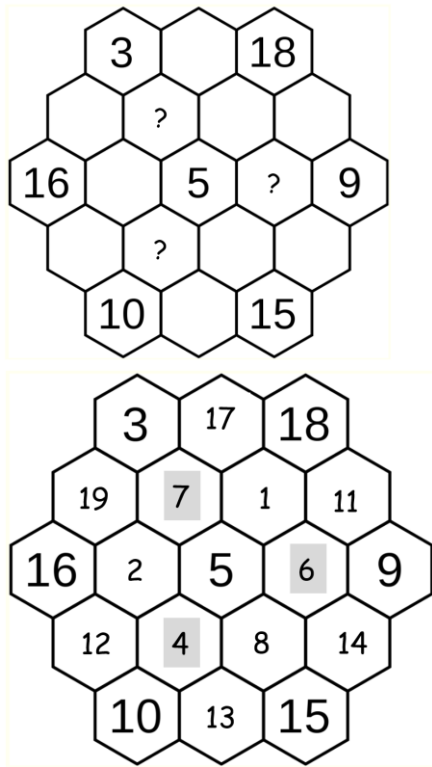
— ● ● ● ● ● ● — ● — ● ● — ●

“Math is Cool” Masters -- 2025-26

5th Grade

Linda Moore Triple Jump Solutions

	Answer	Solution
1	276 [minutes]	Four hours and 36 minutes equals how many minutes? 1 hour = 60 minutes $4 \times 60 + 36 = 276$
2	8	Evaluate: $24 \div (4 \times 2 - 5)$ $24/3 = 8$
3	216 [cm ³]	In cubic centimeters, what is the volume of a rectangular prism that has a base area of 48 square centimeters and a height of $4\frac{1}{2}$ centimeters? $48 \times 4.5 = 216$
4	2280 [spectators]	The largest stadium in the world is in North Korea. It covers 50 acres and is 197 feet tall, holding up to 114,000 spectators. What is the average number of spectators held per acre of this stadium? $114000/50 = 2280$
5	20 [cubes]	Identical unit cubes are stacked against the walls in a corner of a room. How many of the cubes in the stack are not visible? Bottom row contains: $5+4+3+2+1 = 15$ Next row up contains: $4+3+2+1 = 10$ Next row up contains: $3+2+1 = 6$ Next = 3 Top = 1 Total = 35 Visible = 15 $35 - 15 = 20$ 

6	21 [miles]	Brandon runs along a straight road at a constant rate of 7 miles per hour. Two hours later, Liandro starts riding his bike in the same direction at a constant rate of 21 miles per hour. What is the total distance that Brandon runs, in miles, before Liandro catches up with him? $7(x + 2) = 21x$ $X = 1, 21x = 21$
7	780	What even integer between 700 and 800 is divisible by both 5 and 13? $5 \times 13 = 65$, so the number will be a factor of 65. It's more than 65×10, 65×11 is odd, $65 \times 12 = 789$.
8	17	In this magic hexagon, all of the integers 1 through 19 should be placed one per hexagon so that the sum along every straight line in any direction is always equal to 38. The rows will consist of 3, 4 or 5 hexagons. Some numbers have already been placed. What is the sum of the three numbers in the hexagons that are indicated with a question mark?  $\text{Sum} = 7 + 6 + 4 = 17$

9	$29 [= X + Y]$	The set of integers: 5, 7, 12, X, Y, has the same median, unique mode, and range. What is the sum of X and Y, if $X > Y > 6$? 12 has to be the median in this case, which means that $Y = 12$ to also make it the mode. X would then be 17 to make the range = 12 $12 + 17 = 29$
10	5 [= sum]	A sequence of numbers begins as follows. Starting with the third term, each term is equal to the term preceding it minus the term preceding that one. What is the sum of the first 2026 terms of the sequence? 1, 3, 2, -1, ... 1, 3, 2, -1, -3, -2, 1, 3, 2, ... The sequence repeats itself every 6 terms. The sum of the 6 terms is $1 + 3 + 2 - 1 - 3 - 2 = 0$ $2026/6 = 337 \text{ r}4$ The sum of the first 337 cycles, or $337 \cdot 6 = 2022$ terms, is 0. The remaining 4 terms sum to $1 + 3 + 2 - 1 = 5$.

“Math is Cool” Masters -- 2025-26

5th Grade

College Bowl Round #1 Solutions

	Answer	Solution
1	81	What is nine squared? $9^2 = 81$
2	60 [mph]	A car travels two hundred seventy miles in four and a half hours. What is its average speed in miles per hour? $270 / 4.5 = 60$
3	176 [sq cm]	A rectangle has a perimeter of fifty-four centimeters and a length of sixteen centimeters. What is the area of the rectangle in square centimeters? $w = 27 - 16 = 11$; $A = 16 \times 11 = 176$
4	29	Three consecutive odd integers have a sum of eighty-one. What is the largest of the three integers? $n + (n+2) + (n+4) = 81$; $n = 25$; largest = 29
5	10 [marbles]	A jar contains red and blue marbles in a ratio of three to five. There are forty marbles in total. How many more blue marbles are there than red marbles? $25 - 15 = 10$
6	2112	What is the next palindrome year after the year two thousand two? A palindrome reads the same forwards and backwards. 2112 reversed = 2112
7	15 [minutes]	A pump fills a tank at fifteen gallons per minute, while a drain empties it at nine gallons per minute. The tank holds ninety gallons. Starting empty, how many minutes will it take to fill the tank? $90 / 6 = 15$
8	56	What is the sum of all positive factors of twenty-eight? $1+2+4+7+14 + 28 = 56$
9	5 [numbers]	How many of the following numbers are divisible by eleven: Forty-four, sixty-six, eighty-eight, ninety-nine and one hundred ten. 44, 66, 88, 99, 110 are all divisible by 11
10	365	What is four thousand fifteen divided by eleven? $4015 / 11 = 365$

“Math is Cool” Masters -- 2025-26

5th Grade

College Bowl Round #2 Solutions

	Answer	Solution
1	56	What is the value of seven squared plus seven? $49 + 7 = 56$
2	3 [cups]	A recipe calls for three-quarters of a cup of sugar per batch. How many cups of sugar are needed for four batches? $(3/4) \times 4 = 3$
3	24 [sq units]	A right triangle has legs of length six and eight units. What is its area in square units? $(1/2) \times 6 \times 8 = 24$
4	32	The sum of two numbers is fifty and their difference is fourteen. What is the larger of the two numbers? 32, 18
5	20 [ways]	In how many ways can three students be chosen from a group of six to form a committee? $6C3 = 20$
6	19 [integers]	How many positive integers less than one hundred are divisible by five? 10 multiples of 5 from 5 to 95, and 9 multiples of 10 from 10 to 90.
7	2 [hours]	Train A leaves a station at nine AM traveling at sixty miles per hour. Train B leaves the same station at ten AM traveling in the same direction at ninety miles per hour. How many hours does Train B travel before it catches Train A? At 10 AM, A is 60 miles ahead. Gap closes at 30 mph; $60/30 = 2$ h after 10 AM
8	120 [degrees]	What is the measure of each interior angle of a regular hexagon in degrees? $(6-2) \times 180 / 6 = 120$
9	18 [= gcf]	What is the greatest common factor of seventy-two and ninety? $72 = 8 \times 9$; $90 = 10 \times 9$; GCF = 18
10	16	Twelve times four equals what number times three? $12 \times 4 = 48$ $16 \times 3 = 48$

“Math is Cool” Masters -- 2025-26

5th Grade

College Bowl Round #3 Solutions

	Answer	Solution
1	90	What is the value of one hundred minus the square root of one hundred? $100 - 10 = 90$
2	20,000 [sq cm]	How many square centimeters are in two square meters? 1 square meter = $100 \times 100 = 10000$ square cm $10000 \times 2 = 20,000$ square cm
3	20	Lily counts up by fives. The seventh number she says is forty. What was the third number she said? 10, 15, 20, 25, 30, 35, 40
4	597 [= A + B + C]	If A equals ninety-nine, B equals one hundred ninety-nine and C equals two hundred ninety-nine, what is the value of A plus B plus C? $99 + 199 + 299 = 100 + 200 + 300 - 3 = 600 - 3 = 597$
5	21 [= mean]	The mean of six numbers is fifteen. When two of the numbers are removed, the mean of the remaining four is twelve. What is the mean of the two removed numbers? $90 - 48 = 42$; $42 / 2 = 21$
6	12 [numbers]	The digits one, two, three and four can be used exactly once each to write a certain number of four-digit numbers. How many of these four-digit numbers are even? Last digit: 2 or 4 (2 choices); remaining 3 digits: $3! = 6$ arrangements; $2 \times 6 = 12$
7	11 [cm]	In centimeters, what is the radius of a circle that has a circumference of twenty-two pi centimeters? $C = 22\pi = 2\pi r$ $r = 11$
8	110 [= sum]	What is the sum of the first ten positive even numbers? $2+4+6+8+10+12+14+16+18+20 = 110$
9	72 [= LCM]	What is the least common multiple of eight and eighteen? $8 = 2^3$, $18 = 2 \times 3^2$ $LCM = 2^3 \times 3^2 = 72$
10	50	What is twenty percent of two hundred fifty? $10\% = 25$ $20\% = 2 \times 25 = 50$

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5th Grade

College Bowl Round #4 Solutions

	Answer	Solution
1	17	What is the value of three to the fourth power minus four to the third power? $81 - 64 = 17$
2	4,320 [minutes]	How many minutes are in three days? $3 \times 24 \times 60 = 4320$
3	100 [sq cm]	A square and a regular pentagon have the same perimeter. The side length of the pentagon is eight inches. In square inches, what is the area of the square? Perimeter of pentagon = $8 \times 5 = 40$ Square side length = $40/4 = 10$ Square area = $10^2 = 100$
4	3 [cats]	There are two cats in front of a cat, two cats behind a cat and a cat in the middle. What is the fewest number of cats there could be? With 3 cats, there are 2 in front of the last one, there are 2 behind the first one, and there is one in the middle.
5	6 [numbers]	How many different three-digit numbers can be formed using the digits two, five and eight if each digit is used exactly once? $3! = 6$
6	37 [= 4 th term]	A sequence starts with two, and each term equals double the previous term plus three. What is the fourth term? 2, 7, 17, 37
7	20 [%]	An integer is chosen at random from the integers one through ten inclusive. What is the probability in percent that it contains the digit one? 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 2 out of the 10 contain a '1'. $2/10 = 20\%$
8	36 [cm]	A right triangle has legs of nine and twelve centimeters. What is its perimeter in centimeters? $9+12+15 = 36$
9	5 [numbers]	How many of the following are perfect squares? Sixteen, thirty-six, fifty, eighty-one, one hundred, and one hundred forty-four. 16, 36, 81, 100, 144

10	250 [words]	Lennon can read three thousand seven hundred fifty words in fifteen minutes. On average, how many words does he read per minute? $3750/15 = 250$
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5th Grade

College Bowl Round #5 Solutions

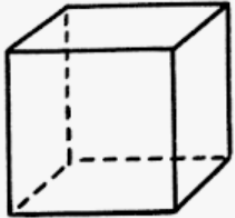
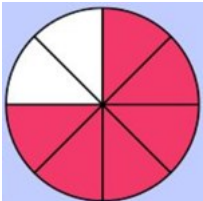
	Answer	Solution
1	63 [quarters]	How many quarters are equal in value to fifteen dollars and seventy-five cents? 1 dollar = 4 quarters 15 dollars = $4 \times 15 = 60$ quarters 75 cents = 3 quarters $60 + 3 = 63$
2	2350 [g]	How many grams are in two point three five kilograms? 1 kg = 1000 g $1000(2.35) = 2350$
3	6 [days]	If it takes five workers twelve days to complete a job, how many days would it take ten workers to complete the same job? $5 \times 12 = 60$; $60 / 10 = 6$
4	81 [\$]	A twenty-five percent discount is applied to a one hundred twenty dollar item, then an additional ten percent loyalty discount is applied. What is the final price in dollars? $120 \times 0.75 = 90$; $90 \times 0.9 = 81$
5	20 [multiples]	How many multiples of one hundred are between one and two thousand twenty-six? $100 \times 20 = 2000$, so there are 20.
6	110 [ways]	Ella has twelve different colored shirts and ten different colored jackets. In how many ways can she pick one shirt and one jacket assuming that she does not want to wear her blue shirt? 12 shirts, but really 11 because no blue. 10 jackets $11 \times 10 = 110$
7	127 [= next term]	A sequence starts with three, seven, fifteen, thirty-one, sixty-three, and so on. What is the next term? Pattern: $x2 + 1$ each time $63 \times 2 + 1 = 127$
8	16 [units]	What is the side length in units of a regular nonagon with a perimeter of one hundred forty-four units? $144 / 9 = 16$

9	14 [= median]	What is the median of the following set of numbers: one, thirty-three, nine hundred ninety-nine, twelve, sixteen, zero Put in order: 0, 1, 12, 16, 33, 999 Median = halfway between 12 and 16 = 14
10	46	In the sequence one hundred, ninety-one, eighty-two, seventy-three and so on, what is the first term less than fifty? 100,91,82,73,64,55,46

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5th Grade

College Bowl Round #6 Solutions

	Answer	Solution
1	148 [marbles]	Brian has twenty marbles, Ernie has three times as many marbles as Brian, and Navani has eight more marbles than Ernie. How many marbles do they have altogether? Brian = 20 Ernie = $3 \times 20 = 60$ Navani = $60 + 8 = 68$ Total = $20 + 60 + 68 = 148$
2	14 [quarts]	How many quarts are in three and a half gallons? 1 gallon = 4 quarts $3.5 \times 4 = 14$
3	12 [edges]	How many edges does a cube have? 
4	6 [slices]	A circle is split into eight equal slices. How many slices have to be colored in to represent three-fourths of the circle? 
5	14 [points] [N]	Seven students scored the following on a quiz: eight, eleven, fourteen, seventeen, twenty, twenty-five, and one unknown score N. If the median is fourteen, what is the largest possible integer value of N? With 7 scores, median = 4th value when sorted. N must be ≤ 14 to keep 14 in 4th place. Largest N = 14.
6	7	What is the largest prime factor of eighty-four? $84 = 2^2 \times 3 \times 7$; largest prime = 7

7	30 [miles]	Two boats leave the same dock at the same time. One travels north at eight miles per hour, the other travels east at six miles per hour. How many miles apart are they after three hours? 24 mi N, 18 mi E; $\sqrt{24^2 + 18^2} = \sqrt{900} = 30$
8	27	What is the value of the square root of one hundred forty-four plus the square root of two hundred twenty-five? $12 + 15 = 27$
9	72	Four over twenty-four equals twelve over what number? $4/24 = (4 \times 3)/(24 \times 3) = 12/72$
10	96 [tiles]	A rectangular tile measures two inches by three inches. What is the smallest number of tiles needed to completely cover a two foot by two foot square region? The square is 24" x 24". $24/2 = 12$, and $24/3 = 8$. $12 \times 8 = 96$